

EtherCAT Interface Software: Quick Start Guide

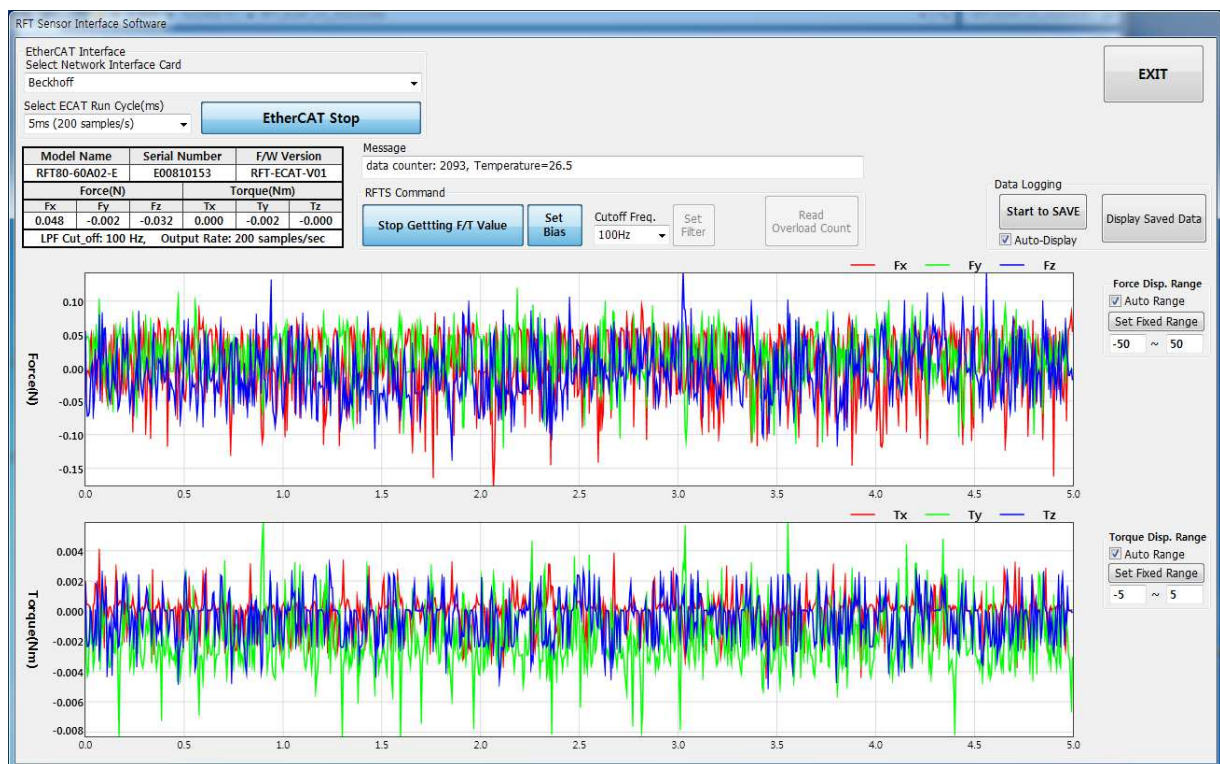
1. Overview

The RFT Series Sensor Interface Software is designed to monitor and control force/torque sensors via EtherCAT communication. It provides real-time data visualization, sensor configuration, and logging capabilities.

2. System Requirements

- 1) Windows-based PC
- 2) Ethernet Network Interface Card (NIC)
- 3) RFT Series Force/Torque Sensor (EtherCAT version)

3. Main Interface Description



1) EtherCAT Interface Setup

- * Select Network Interface Card
- * Choose the NIC connected to the EtherCAT network.
- * Select **ECAT Run Cycle** (ms). [Example: 5 ms (200 samples/sec)]
- * **Start ECAT** to Initialize EtherCAT communication with the sensor.

2) Sensor Information Panel

Once communication is established, this information is automatically updated.

- * Model Name | Serial Number | Firmware Version

3) Real-Time Data Display

Once data acquisition starts, the measured values are automatically updated in real time.

- * Real-time force/torque data: Fx, Fy, Fz, Tx, Ty, Tz
- * Upper graph: Force (N)
- * Lower graph: Torque (Nm)

4) RFTS Command Panel

The RFTS Command Panel provides control and configuration functions for the sensor.

- * **Get F/T Value Continuously:**
Starts/stops real-time streaming of force/torque data (Toggle)
- * **Set Bias:**
Toggles zero setting (offset removal) on/off
- * **Cut-off Frequency:**
Selects LPF cut-off frequency [e.g., 1Hz ~ 1kHz(no filter)]
- * **Set Filter:**
Applies the selected cutoff frequency
- * **Read Overload Count:**
Displays the number of overload events detected by the sensor

5) Data Logging

- * **Start to SAVE** to begin recording sensor data (Toggle)
- * Auto-Display (✓): automatically shows logged data after logging
- * Display Saved Data: opens previously recorded data for review

6) Display Range Settings

- * Force / Torque Display Range: Adjustable (Auto/Manual)
- * Auto Range (✓): automatically scales to fit input data
- * Set Fixed Range: Displays data within the set range only

7) Exit

- * Click EXIT to safely close the application

4. Basic Operation Procedure

- ➊ Connect the sensor to the PC via EtherCAT
- ➋ Launch the software
- ➌ Select the correct Network Interface Card
- ➍ Set the desired ECAT Run Cycle
- ➎ Click Start ECAT
- ➏ Verify that sensor information is displayed
- ➐ Click Get F/T Value Continuously to start measurement
- ➑ (Optional) Click Set Bias to zero the sensor
- ➒ Adjust filter and display settings as needed
- ➓ Use Start to SAVE for data logging